

Explainability

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 24: Responsible AI

1. Explainability vs. Interpretability — A Critical Distinction

These terms are related but describe different properties of AI systems, and conflating them leads to poor architecture decisions.

Property	Definition	Examples	Key Implication
Interpretability	The degree to which a model's internal mechanisms directly correspond to human-understandable logic — an intrinsic property of model architecture	Linear regression (high), decision trees (moderate), deep neural networks (low)	Interpretable models are often preferable for regulated high-stakes contexts where auditability of the model itself is required.
Explainability	Post-hoc methods that provide explanations for model outputs regardless of model architecture — applied after the fact to any model	LIME, SHAP, counterfactuals, saliency maps	Complex models can still be explainable with appropriate tools, but explanations are approximations, not the model's actual reasoning process.

The practical implication: for regulated high-stakes contexts, interpretable models are preferable when accuracy permits. When complex models are necessary for accuracy, explainability tools become essential compensating controls. Neither interpretability nor explainability alone satisfies all regulatory requirements — the selection of approach requires understanding both the regulatory standard and the technical constraints.

2. Why Explainability Matters — Legal, Ethical, Technical

Explainability requirements arise from three distinct and independently compelling sources:

- **Legal:** GDPR Article 22 provides a right to 'meaningful information about the logic involved' in automated decisions with significant effects. The EU AI Act requires high-risk AI outputs to be interpretable enough to enable human oversight. US federal agencies under OMB guidance must ensure AI used in consequential decisions is explainable. CFPB and OCC require that adverse action notices for credit decisions provide specific reasons for denial.
- **Ethical:** Individuals who receive adverse AI decisions — loan denials, job rejections, parole conditions — have a legitimate interest in understanding why and in having a basis to contest incorrect decisions. Unexplained adverse decisions undermine autonomy and due process.
- **Technical:** Explainability tools expose how models actually make decisions, revealing biases, spurious correlations, and brittleness that are invisible in aggregate accuracy metrics. A model with 97%

accuracy on a test set may be relying on a spurious correlation invisible without explainability analysis.

Exam Tip: Explainability is simultaneously a legal compliance mechanism, an ethical practice, and a quality assurance tool. When an exam question frames explainability as purely ethical, consider whether legal or technical dimensions are the more precise answer.

3. LIME — Local Interpretable Model-Agnostic Explanations

LIME (Ribeiro et al., 2016) explains individual predictions by approximating the complex model locally with a simple interpretable surrogate. For a specific prediction, LIME perturbs the input in small ways, collects the model's outputs for those perturbed inputs, and fits a simple linear model to approximate the relationship between features and outputs in that local neighborhood.

Key properties of LIME:

- **Model-agnostic:** Works for any model type — image classifiers, text classifiers, tabular models — because it treats the model as a black box and probes it externally.
- **Local, not global:** LIME explains this specific prediction, not the model overall. The local linear approximation may not accurately represent model behavior elsewhere in the input space.
- **Actionable outputs:** A credit decision system flagging high risk can be explained via LIME as: 'For this specific applicant, the three features most strongly driving the high-risk score are employment tenure, recent credit inquiries, and debt-to-income ratio.' This tells the applicant what drove the decision.

LIME's limitation is stability: small changes to the input can produce different explanations, which creates challenges for audit consistency. When LIME explanations are used in compliance contexts, variance testing is a necessary quality control.

4. SHAP — Shapley Additive Explanations

SHAP (Lundberg and Lee, 2017) provides feature attribution grounded in cooperative game theory. Each feature receives a Shapley value — computed by averaging its marginal contribution to the prediction across all possible orderings of features. SHAP values are additive: they sum exactly to the model's output, providing mathematically rigorous attribution.

SHAP capabilities at both scales:

Scale	What It Provides	Typical Use
Local	Individual SHAP values for a specific prediction showing how much each feature contributed to reaching that specific output	Explaining individual credit decisions, fraud flags, or hiring recommendations to affected individuals or auditors
Global	Aggregate SHAP values across all predictions showing overall feature importance and direction of influence	Model development quality assurance, bias detection, regulatory documentation of what the model relies on

- **TreeSHAP:** Provides exact Shapley values for tree-based models — XGBoost, LightGBM, RandomForest — in polynomial time, making it fast enough for production inference logging.
- **DeepSHAP:** Approximates Shapley values for deep learning models using a combination of DeepLIFT and SHAP framework.

SHAP is the most widely adopted explainability library in production ML. Know it for the exam: SHAP provides both local and global explanations, is grounded in game theory, and SHAP values sum to the model's output.

5. Counterfactual Explanations — Actionable Recourse

Counterfactual explanations answer a different question from LIME or SHAP: not 'why did this decision occur?' but 'what would need to change for the decision to be different?'

Example: A loan applicant denied credit receives: 'If your annual income were \$10,000 higher and you had no missed payments in the last 12 months, this application would likely be approved.' This is actionable — the applicant understands specific, potentially achievable changes that would change the outcome.

Why counterfactuals are particularly valuable in regulated contexts:

- Legally meaningful recourse: counterfactuals satisfy the 'what could change' dimension of adverse action notices without requiring disclosure of model internals.
- Privacy-preserving: counterfactuals can be generated without revealing how other individuals' data influenced the model.
- Accessible to non-technical stakeholders: the format corresponds to how people naturally reason about decisions.
- **DiCE (Microsoft Research):** Diverse Counterfactual Explanations — generates multiple diverse counterfactuals, giving individuals options for different paths to a positive outcome.

Key Insight: Counterfactual explanations are the format closest to how people reason about decisions and are particularly suited for satisfying legal recourse requirements while preserving model confidentiality and individual privacy.

6. Attention and Saliency for Neural Networks

For neural network models — particularly transformers and convolutional networks — architecture-specific methods visualize which parts of the input influenced the prediction.

- **Attention Weights:** In transformer models, attention weights indicate which tokens or image patches the model weighted heavily when generating a prediction. Useful for understanding which input segments influenced LLM outputs.
- **Saliency Maps:** Visualize which pixels most influenced a classification decision in image models. Gradient-based methods compute the sensitivity of the output to each input pixel.
- **Grad-CAM (Gradient-weighted Class Activation Mapping):** Produces heatmaps highlighting image regions most important for a predicted class, using gradients from the predicted class flowing into the final convolutional layer.

Critical caveat: attention weights do not perfectly correspond to causal importance. Saliency maps can highlight spurious correlations — a famous example is an image classifier that classified wolves correctly by detecting snow in the background rather than wolf features, producing confident but misleading saliency maps. These methods generate hypotheses that require verification through additional analysis, not conclusions.

7. Global vs. Local Explainability — Both Are Required

Scale	Purpose	Methods	When Used
Global	Describes how the model makes decisions in general — which features matter across all predictions, how behavior varies across the input space	SHAP summary plots, permutation feature importance, partial dependence plots	Model development QA; pre-deployment bias review; regulatory documentation; ongoing monitoring
Local	Describes how the model made a specific prediction — for this individual, why this output	LIME, individual SHAP values, counterfactuals, attention weights for the specific input	Adverse action notices; auditor review of specific decisions; customer-facing explanations; dispute resolution

Both scales are necessary. Global explainability is essential for model development — revealing that a model relies on a protected attribute proxy exposes a systematic bias problem. Local explainability is essential for operational accountability — enabling individuals to understand and contest specific decisions. Deploying only one is insufficient for comprehensive compliance.

8. Explainability Enables Fairness Assessment

Explainability tools directly operationalize fairness evaluation at the individual level, complementing group-level fairness metrics.

- SHAP disaggregated analysis reveals whether different features drive predictions for different demographic groups — which can indicate bias even when group-level approval rates appear similar.
- Counterfactual fairness testing reveals disparate recourse: do minority applicants need to change more or harder-to-change attributes to receive a positive outcome? If so, the model is more hostile to minority applicants even when approval rates are equal.
- Saliency maps can reveal when a model classifies different demographic groups by different visual features — potentially indicating reliance on demographic proxies rather than task-relevant features.

If a hiring model uses educational institution as a strong feature for majority-group applicants but relies primarily on years of experience for minority-group applicants, the model is treating groups differently. This is only discoverable through disaggregated explainability analysis.

9. Explainability for LLMs — A Harder Problem

Explainability for large language models is substantially harder than for traditional ML because the computational process is distributed across billions of parameters and hundreds of attention heads with no clean correspondence to human-interpretable features.

- Attention visualization is possible but limited — no single attention pattern corresponds neatly to why a specific output was generated. Attention distribution can be informative but is not a complete explanation.
- **Chain-of-Thought Prompting:** Asking the LLM to explain its reasoning step by step improves understandability of outputs. However, the expressed reasoning does not guarantee it reflects the model's actual computational process — LLMs can confabulate plausible-sounding but inaccurate explanations of their own outputs.
- **Constitutional AI and RLHF (Anthropic):** Shape model behavior toward more consistent and verifiable reasoning, improving the reliability of chain-of-thought explanations, without making the underlying computation transparent.

For LLM deployment in high-stakes contexts, explainability approaches focus on output inspection — does this output make sense, is it consistent with policy, does it contradict known facts — rather than computational transparency. This represents a fundamental difference from traditional ML explainability.

Real-World Incident — MIT Pneumonia Detection Model (2020): Researchers at MIT developed a deep learning model matching radiologist accuracy in detecting pneumonia from chest X-rays. When asked to explain flagged X-rays, the model could not provide medically interpretable reasoning. It was identifying pixel patterns correlated with but not corresponding to known medical features. Clinicians could not integrate unexplained AI predictions into clinical reasoning; regulators could not evaluate safety; patients could not contest the basis of diagnoses. A highly accurate model that cannot explain itself is, in many clinical contexts, a model that cannot be used.

10. Security Perspective — Exploiting Explainability Systems

Explainability systems themselves present adversarial attack surfaces that practitioners must anticipate.

- **Scaffold Attacks:** Adversaries learn which features the explainability system identifies as important, then craft inputs that change the apparent important features without changing the actual correlation with the target. This defeats audit systems examining feature importance for bias, because the explanation changes while the underlying discrimination persists.
- **Explanation Inconsistency Attacks:** When an adversary understands the explainability method being used, they construct inputs where the explanation and actual decision driver diverge — the model classifies correctly for the wrong reason, making the explanation misleading to auditors.
- **Compliance Gaming:** Organizations select explainability methods that show favorable feature attributions while the model uses other, unexplained features for actual decisions. This is the organizational equivalent of explanation inconsistency — deliberate rather than adversarial.

Robust explainability requires using multiple methods and cross-checking for consistency. If LIME, SHAP, and counterfactual analysis produce substantially different feature attributions for the same prediction, that inconsistency is itself a signal requiring investigation.

11. Regulatory Requirements for Explainability

Multiple regulatory frameworks create overlapping explainability requirements:

- GDPR Article 22: 'meaningful information about the logic involved' in automated decisions with significant effects on individuals.
- EU AI Act: high-risk AI must provide outputs 'interpretable by the deployers to enable human oversight' — the deployer must be able to understand enough to perform meaningful review.
- CFPB and OCC adverse action notices: credit decision AI must generate specific reasons for denial that can be communicated to applicants.
- FDA on AI-based Software as a Medical Device (SaMD): emphasizes that AI recommendations should be interpretable by clinicians before being acted upon.
- NIST AI RMF EXPLAIN function: explicitly addresses explainability as a core AI risk management capability across the AI system lifecycle.

For organizations operating across jurisdictions, the most demanding applicable explainability requirement typically sets the practical standard. A credit model deployed in the EU must satisfy both GDPR logic disclosure requirements and EU AI Act deployer interpretability requirements simultaneously.

Key Terms Glossary

Term	Definition
Interpretability	Intrinsic property of a model — how directly its internal mechanisms correspond to human-understandable logic. Linear models are interpretable; deep neural networks are not.
Explainability	Post-hoc methods providing explanations for model outputs regardless of model architecture — applied externally to any model type.
LIME	Local Interpretable Model-Agnostic Explanations — explains individual predictions by fitting a local linear surrogate to perturbed inputs near the prediction point.
SHAP	Shapley Additive Explanations — game-theoretic feature attribution providing both local and global explanations; SHAP values sum to the model output.
TreeSHAP	Exact SHAP computation for tree-based models (XGBoost, LightGBM, RandomForest) in polynomial time, suitable for production logging.
Counterfactual Explanation	Explains what would need to change about an input for the decision to be different; provides actionable recourse information.
DiCE	Diverse Counterfactual Explanations (Microsoft Research) — generates multiple diverse counterfactuals for a single prediction.
Grad-CAM	Gradient-weighted Class Activation Mapping — produces heatmaps highlighting image regions important for a CNN classification.
Saliency Map	Visualization showing which input pixels or features most influenced a neural network prediction.
Global Explainability	Describes model behavior in general — overall feature importance

	across all predictions.
Local Explainability	Describes how the model made a specific prediction — for this individual, why this output.
NIST AI RMF EXPLAIN	AI Risk Management Framework function addressing explainability as a core AI risk management capability.

Quick Revision Checklist

- Interpretability is a model property (how transparent the model is internally); explainability is post-hoc (explaining outputs of any model externally).
- LIME: local, model-agnostic, approximates model behavior near a specific prediction with a simple surrogate; variance is a limitation.
- SHAP: game-theoretic feature attribution; provides both local and global explanations; SHAP values sum to the model output; TreeSHAP is production-speed for tree models.
- Counterfactuals: answer 'what would need to change?'; provide actionable recourse; privacy-preserving; DiCE generates diverse counterfactuals.
- Attention/saliency/Grad-CAM: neural network-specific; generate hypotheses, not conclusions — can highlight spurious correlations.
- Global explainability: for model development QA and regulatory documentation. Local explainability: for adverse action notices and individual accountability.
- Explainability enables fairness assessment: SHAP disaggregated analysis reveals differential feature use; counterfactual testing reveals disparate recourse.
- LLM explainability is harder: chain-of-thought improves understandability but confabulation means expressed reasoning may not reflect actual computation.
- MIT pneumonia model (2020): accurate but unexplainable model that clinicians could not integrate into practice — accuracy without explainability is insufficient in high-stakes contexts.
- Security: scaffold attacks, explanation inconsistency attacks, and compliance gaming all exploit explainability systems — multiple methods and cross-checking are required.
- Regulatory basis: GDPR Art 22, EU AI Act EXPLAIN, CFPB/OCC adverse action, FDA SaMD guidance, NIST AI RMF EXPLAIN function.

10 Exam-Based MCQs — Explainability

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. *Scenario:* A bank uses an ML model for mortgage application decisions. An applicant who was denied credit contacts the bank to understand why and what they can do to improve their chances. The compliance officer needs to provide a meaningful explanation. A compliance officer needs to explain to a loan applicant why their application was denied and what they could do to receive approval in the future. Which explainability method BEST serves this requirement?

- A) SHAP global summary plot showing overall feature importance across all applicants
- B) Counterfactual explanation specifying which attribute changes would result in approval

- C) Grad-CAM heatmap highlighting the most influential regions of the application form
- D) LIME global explanation of how the model treats employment tenure across all predictions

Answer: B **Explanation:** B is correct — counterfactual explanations directly answer 'what would need to change for the decision to be different?' This provides the actionable recourse information most useful to an individual applicant and most aligned with legal adverse action notice requirements. A is wrong because a global SHAP summary describes overall model behavior across all applicants, not why this specific application was denied. C is wrong because Grad-CAM applies to image models and produces visual heatmaps, not the structured explanations needed for a credit decision explanation to an individual. D is wrong because a LIME global explanation describes model behavior generally, not local explanation for the specific prediction.

Q2. Scenario: An AI system is used to screen job applications. An external audit team has been asked to determine whether the model is applying different decision criteria to candidates from different demographic groups. A security audit reveals that an AI hiring system uses different features to score candidates from different demographic groups. Which tool MOST directly reveals this type of differential model behavior?

- A) LIME local explanations for five randomly selected rejected applications
- B) SHAP disaggregated analysis examining feature attribution patterns by demographic group
- C) Grad-CAM applied to candidate resume images
- D) Counterfactual explanations for the top-scoring candidates in each group

Answer: B **Explanation:** B is correct — SHAP disaggregated analysis computes feature attributions separately for different demographic groups, directly revealing whether different features drive predictions for different groups. This is the analytical approach designed to surface differential model behavior across protected categories. A is wrong because five individual LIME explanations cannot establish a pattern across demographic groups — this is sample analysis, not disaggregated systematic analysis. C is wrong because Grad-CAM applies to convolutional neural networks processing images, not typical tabular or text-based hiring models. D is wrong because counterfactual explanations for top-scoring candidates reveal what those individuals could change, not whether the model treats different groups differently.

Q3. Scenario: A financial institution's gradient-boosted tree fraud detection model is undergoing regulatory examination. Auditors want to verify the model is not relying on proxies for protected characteristics such as race or national origin. A risk manager asks the data science team to provide an explanation of an AI fraud detection model that regulatory auditors can examine to verify the model is not relying on prohibited features. The model is a gradient-boosted tree ensemble. Which approach BEST satisfies this requirement?

- A) Provide LIME explanations for five individual fraud cases
- B) Provide SHAP global feature importance summary with TreeSHAP values across all predictions
- C) Retrain the model as a logistic regression to make it interpretable
- D) Provide attention visualizations showing which input tokens influenced the model

Answer: B **Explanation:** B is correct — SHAP global feature importance with TreeSHAP provides exact,

mathematically rigorous feature attributions for tree-based models (gradient-boosted trees are in scope for TreeSHAP) at production speed. A global summary across all predictions shows auditors what the model relies on systematically, which is what regulatory review of prohibited feature usage requires. A is wrong because five individual LIME explanations address local behavior, not the systematic feature reliance pattern auditors need to assess. C is wrong because retraining as logistic regression introduces a new, different model with different behavior — it doesn't explain the original model and may sacrifice accuracy. D is wrong because attention visualizations apply to transformer models, not tree ensembles.

Q4. An image classification model correctly identifies 95% of images in its test set. Saliency maps reveal the model identifies 'cats' by detecting background furniture patterns rather than cat features. What does this finding MOST directly indicate?

- A) The model is performing adversarial attacks on the test set
- B) The model has learned a spurious correlation that will cause poor generalization to new environments
- C) The saliency maps are malfunctioning — accurate models cannot rely on spurious features
- D) The model should be retrained with a smaller dataset to reduce overfitting

Answer: B **Explanation:** B is correct — this is the classic spurious correlation problem revealed by explainability analysis. The model learned that furniture patterns in photos correlate with cats in the training dataset (because cat photos are taken indoors), but this correlation will not generalize to cats in other environments. High test accuracy does not prevent this — if the test set has the same distribution bias, the spurious feature will produce accurate predictions on the test set while failing in deployment. A is wrong because the model is not performing adversarial attacks — it is relying on a spurious learned feature. C is wrong because saliency maps functioning correctly can absolutely reveal spurious feature reliance — this is precisely their QA value. D is wrong because dataset size is not the primary issue; the training data distribution is biased in a way that creates a spurious correlation.

Q5. Which statement BEST describes what SHAP values mathematically guarantee about feature attribution?

- A) SHAP values represent the causal effect of each feature on the model output
- B) SHAP values sum to the model's output, providing additive feature attributions grounded in game theory
- C) SHAP values guarantee the model is not using prohibited features
- D) SHAP values provide exact explanations for deep learning models at production speed

Answer: B **Explanation:** B is correct — the defining mathematical property of SHAP is additivity: SHAP values for all features sum exactly to the model's prediction (relative to a baseline). This is derived from the Shapley value axioms in cooperative game theory. A is wrong because SHAP values represent associative feature contribution, not causal effect — a feature with a high SHAP value may be correlated with the target through other variables, not causally responsible. C is wrong because SHAP values show what features the model uses; they cannot guarantee compliance with feature restrictions — a model can have high SHAP values for prohibited features. D is wrong because DeepSHAP provides approximations for deep learning, not exact values; TreeSHAP provides exact values, but only for tree-based models.

Q6. GDPR Article 22 requires 'meaningful information about the logic involved' for automated decisions. Which explainability output MOST directly satisfies this requirement for an individual affected by an automated credit decision?

- A) A technical paper describing the neural network architecture used in the credit model
- B) A local SHAP explanation or counterfactual showing which factors drove this specific individual's decision
- C) A global feature importance chart showing overall model behavior across all credit decisions
- D) A statement that the model was trained on historical credit data with 94% accuracy

Answer: B **Explanation:** B is correct — GDPR Article 22 requires meaningful information to the specific individual affected by the decision about the logic that produced their specific outcome. Local SHAP values or a counterfactual explanation directly address this by explaining what drove this person's specific decision. A is wrong because a technical architecture paper describes the model in general, not the logic applied to this individual's specific data. C is wrong because global feature importance describes aggregate model behavior, not the logic applied to the specific individual's decision. D is wrong because an accuracy statement and training data description convey nothing meaningful about why this individual's application was decided as it was.

Q7. A practitioner asks an LLM to explain its reasoning for a specific response by providing a chain-of-thought explanation. The explanation sounds plausible and internally consistent. Which important limitation must the practitioner recognize?

- A) Chain-of-thought explanations are only valid for transformer models smaller than 7 billion parameters
- B) The expressed chain-of-thought reasoning may not reflect the model's actual computational process — LLMs can generate plausible-sounding but inaccurate self-explanations
- C) Chain-of-thought explanations cannot be used in regulated contexts because they are not auditable
- D) Chain-of-thought outputs should be disregarded entirely because explainability is impossible for LLMs

Answer: B **Explanation:** B is correct — LLM confabulation extends to self-explanation. A model can generate a fluent, internally consistent chain-of-thought narrative that does not accurately reflect the computational path that produced its output. This is the core limitation practitioners must understand: apparent reasoning transparency from chain-of-thought is an improvement over no explanation but is not computational transparency. A is wrong because this limitation applies to LLMs generally, not just specific size ranges. C is wrong because chain-of-thought explanations can be used in regulated contexts as part of an explainability approach, with appropriate acknowledgment of limitations. D is wrong because chain-of-thought provides useful, if imperfect, explanatory signal — treating it as entirely worthless overstates the limitation.

Q8. An organization selects an explainability method that consistently shows favorable feature attributions to regulatory auditors, while the model actually uses different, non-disclosed features for its real decisions. Which attack or risk category does this represent?

- A) Scaffold attack — adversaries have manipulated the explainability system
- B) Explanation inconsistency attack — an external attacker is causing explanation divergence
- C) Compliance gaming — the organization is deliberately selecting explainability methods to produce misleading audit results
- D) Model inversion — the explainability method is exposing training data to auditors

Answer: C **Explanation:** C is correct — compliance gaming is the deliberate organizational behavior of selecting or configuring explainability methods to show favorable attributions while the model uses different features for actual decisions. This is an internal governance failure, not an external attack. A is wrong because scaffold attacks involve external adversaries crafting inputs to change apparent feature importance without changing actual decision drivers. B is wrong because explanation inconsistency attacks involve external adversaries constructing inputs where the explanation and actual driver diverge — the behavior described is deliberate organizational selection, not external attack. D is wrong because model inversion involves reconstructing training data from model parameters, not a relationship between explanation methods and audit outcomes.

Q9. The NIST AI RMF EXPLAIN function addresses which core AI risk management capability?

- A) Documenting who is responsible for AI system decisions and failures
- B) Explainability — providing meaningful explanations of AI system decisions and behaviors to relevant stakeholders
- C) Encrypting AI model parameters to prevent unauthorized model theft
- D) Evaluating AI system performance against fairness metrics across demographic groups

Answer: B **Explanation:** B is correct — the NIST AI RMF EXPLAIN function is one of the framework's core functions and specifically addresses explainability: the ability to provide meaningful explanations of AI decisions, behaviors, and outputs to stakeholders including affected individuals, operators, and oversight bodies. A is wrong because accountability governance is addressed by the GOVERN function in NIST AI RMF, not EXPLAIN. C is wrong because model parameter encryption is a security control addressed in risk management contexts, not the EXPLAIN function. D is wrong because fairness metric evaluation is addressed in the MEASURE function of NIST AI RMF, not EXPLAIN.

Q10. A healthcare AI system correctly flags 97% of pneumonia cases in clinical trials. Clinicians cannot determine what features the model is using to make its flags. Regulators cannot evaluate whether the model's reasoning is medically sound. What does this scenario demonstrate?

- A) High accuracy is sufficient for clinical deployment — explainability is a secondary concern that can be addressed post-launch
- B) The model should be replaced with a lower-accuracy interpretable model regardless of clinical implications

- C) Accuracy without explainability is insufficient for high-stakes deployment — explainability is a prerequisite for clinical integration, regulatory approval, and patient recourse
- D) Explainability requirements do not apply to healthcare AI systems that have been validated on clinical trial data

Answer: C **Explanation:** C is correct — the MIT pneumonia detection model scenario illustrates exactly this: a high-accuracy model that clinicians cannot integrate into clinical reasoning, regulators cannot evaluate for safety, and patients cannot contest is not deployable in most clinical contexts regardless of its accuracy. Explainability is required for clinical reasoning integration, for regulatory safety evaluation, and for patient rights. A is wrong because post-launch explainability implementation is not equivalent to pre-deployment explainability — the FDA and clinical practice require that AI recommendations be interpretable before acting on them. B is wrong because the answer does not require replacing with a lower-accuracy model — it requires that explainability be a design requirement addressed before deployment, potentially through post-hoc methods. D is wrong because FDA guidance on AI-based SaMD explicitly emphasizes interpretability for clinical AI, and clinical trial validation does not substitute for explainability.